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Information Collection on the Legal and Policy Aspects of Space Traffic (Management)

Contribution by Germany

Germany welcomes that COPUOS requested UNOOSA to collect from member states and permanent observers information on the relevant regulations, policies, best practices and requirements at national and regional levels, including for the identification of existing international norms for consideration by the Legal Subcommittee at its sixty-fifth session under the agenda item entitled "General exchange of views on the legal aspects of space traffic management", in order to discuss and decide on a possible way forward for future work relating to the topic (UN Doc. A/80/20, para. 156).

The question of how the international community ensures a future framework for safe and sustainable space traffic is among the most crucial questions of international space governance. Over the past decade, the number of satellites launched has increased exponentially, bringing the number of active and inactive spacecraft in orbit to over 15,000.[1] *Space traffic*, referring to the movement of all space objects, including parts thereof and debris, during all phases of spaceflight, has therefore increased significantly, especially in Earth's orbit. This is inherently linked to the fact that the potential and benefits of space activities for humanity have never been greater.

Over the past decade, Germany has supported initiatives for space traffic coordination and management both at the global and the regional level. As previously stated under the relevant agenda item at the Legal Subcommittee, *Space Traffic Management* can be understood as a coherent set of technical and regulatory provisions for promoting safe access into outer space, operations in outer space and return from outer space to Earth free from physical or radio-frequency interference.[2]

1. Relevant regulations, policies, best practices and requirements at the national and regional levels

Since 2025, the Ministry of Research, Technology and Space is responsible for actively shaping German space policy together with science, industry and European and international partners. Germany is engaged in designing space-related policies and regulations including with regard to space traffic on the national, regional and international level. The orientation toward the principle of sustainability is one of the overarching guidelines of the Federal Government's Space Strategy of 2023. Space Traffic Management was identified as a key topic of the Strategy, including a commitment to help shape an international framework within the United Nations. To support and complement this approach, Germany further supports the regional "EU approach to space traffic management" and is actively engaged in the European Union Space Surveillance and Tracking (EUSST), where Germany and other EU Member States share surveillance and tracking (SST)

[1] A Year of Transition, UNOOSA Annual Report 2023 and OOSA Online Index of Space Objects.
[2] IAA Cosmic Study on Space Traffic Management 2017.

data with each other, process this data, and provide operational services for collision avoidance, re-entry and fragmentation events to users worldwide.

Germany supports the objective articulated in the Guidelines for the Long-Term Sustainability of Outer Space Activities (LTS Guidelines), namely to "promote regulations and policies that support the idea of minimizing the impacts of human activities on Earth as well as on the outer space environment" (Guideline A.2 para 2 lit. d) and views the existing 21 Guidelines as a building block for future international space traffic management. Thus, Germany has reported to the Scientific and Technical Subcommittee on its national implementation of the Guidelines (see UN Doc. A/AC.105/C.1/2023/CRP.9). The summarized practices contribute to the safe coordination of space traffic. For instance, the German Space Situational Awareness Centre (GSSAC) is tasked with the creation of operational SSA information and the provision of operational services (e. g. collision avoidance, re-entry and fragmentation analysis) for governmental recipients. The German Space Agency is currently updating its Product Assurance, Safety and Sustainability Requirements for DLR Space Projects, which are applicable to national missions as well as non-governmental missions receiving financial support from the National Space Program on Innovation and International Cooperation. These missions have to implement specific requirements, including for the mitigation of space debris and other elements of regulating space traffic like launch collision avoidance analyses, one of the technical building blocks already contained in the LTS Guidelines. In an effort to ensure the safe and sustainable use of outer space, Germany is an active member at the Inter-Agency Space Debris Coordination Committee (IADC), a signatory to the ESA Zero Debris Charter and participates in the development of missions and services within the European Space Agency (ESA) Space Safety Programme.

2. Identification of existing international norms

The five UN space treaties are the cornerstone of the international legal framework for activities in outer space and provide important guidance when discussing legal and policy aspects of space traffic (management). First and foremost, Art. I of the Outer Space Treaty provides that the exploration and use of outer space shall be the province of all (hu)mankind and outer space shall be free for exploration and use by all States without discrimination of any kind. The freedom to use and explore outer space is to be enjoyed by all Treaty Parties and finds its limits in the principles of the subsequent articles of the OST. This entails an obligation to use outer space responsibly. This can most appropriately be achieved through international management of space traffic, referring to the entirety of coordinated international efforts, policies, legal regulation, standards, technologies and strategies designed to ensure the safe, sustainable and efficient operation of objects in space, and in particular in Earth orbit. Additional articles of the Outer Space Treaty are relevant: Art. VIII allocates jurisdiction and control and clarifies ownership of objects launched into outer space; according to Art. IX States are to pay due regard to the corresponding interests of all other States Parties and hold international consultations in cases of potential interferences; Art. X addresses opportunities to observe the flight of space objects launched by other States on a basis of equality and with Art. XI States agree to share information on their space activities to the greatest extent feasible and practicable. Internationally accepted rules of the road can also be seen as a prerequisite for an effective international liability regime in accordance with the Liability Convention.

3. Considerations on the way-forward for future work relating to the topic at the COPUOS Legal Subcommittee

The ever-growing number of launches and space objects challenges the safety and long-term sustainability of space activities, including the realisation of economic potentials. Germany will continue to promote ecologically, economically and socially sustainable activities, innovations and business models in its space activities that serve the people and our planet at national, European and international level, in particular within the framework of the United Nations. We continue to advocate for a multilateral approach as well as capacity-building for the long-term sustainable use of space. Germany is already supporting a wide range of capacity-building measures and is endeavouring to further develop this support. We remain keen to facilitate and engage in a dialogue among developing countries, emerging and experienced space faring nations on sustainable space activities to ensure that space continues to serve the benefit of all humankind long into the future.

The importance of strengthened global governance of space activities was recognized by world leaders at the Summit of the Future in Action 56 of the Pact for the Future, which contained a decision to discuss the establishment of a new framework for space traffic. The international space community needs to initiate meaningful discussions on the scientific, technical, legal and policy aspects of space traffic. Germany is convinced that effective space traffic management contributes to the protection of operational space systems, ensures the viability of private and public investments in and benefits from space and plays an important role for a sustainable space environment that benefits all. Therefore, Germany welcomes the establishment of the Expert Group on Space Situational Awareness and is actively contributing to its work. In order to address this urgent issue, COPUOS must rely on the expertise and capacity of both Subcommittees. Therefore, we believe that establishing work streams in both Subcommittees in a timely manner is essential.

We consider it crucial to start substantial discussions in the Legal Subcommittee with a view to reaching consensus on underlying principles of the international framework to ensure the freedom to use and explore outer space. A possible way forward would be to discuss under the existing agenda item or possibly in a dedicated Working Group:

- examples of national approaches to space traffic coordination and management (not technical) as well as relevant regulatory best practices,
- elements to be considered when drafting national regulations,
- participation by non-governmental actors in space traffic coordination/management,
- possible contributions by COPUOS (recommendations, guidelines, etc.) and
- ways and means to promote a common understanding of space traffic management, including through capacity building measures in the field of space law and policy through international cooperation.